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Adaptability Evaluation of Combined Secondary and Tertiary CO_2 Gravity-Stabilized Flooding in Tight Oil Reservoirs

Jin Chen, State Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, Beijing, China / School of Mining and Petroleum Engineering, University of Alberta, Edmonton, Alberta, Canada; Liang Zhao, PetroChina Research Institute of Petroleum Exploration & Development, Beijing, China; Yang Song, Daigang Wang, Kaoping Song, and Xinxuan Qi, State Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum, Beijing, China

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Abstract

Tight oil reservoirs are characterized by significant pore-throat heterogeneity, strong reservoir heterogeneity, and poor water injectivity, which substantially constrain the recovery efficiency of conventional waterflooding. The combined approach of secondary fine waterflooding and CO_2 gravity-stabilized flooding can effectively expand the swept volume, improve fluid flow conditions, and further enhance oil recovery. However, existing adaptability evaluation methods predominantly rely on single experimental or simulation techniques, often resulting in deviations from actual field performance. To address this limitation, this study proposes a comprehensive adaptability evaluation methodology for combined secondary and tertiary CO_2 gravity-stabilized flooding in tight oil reservoirs. A comprehensive dynamic analysis was conducted to assess the performance of secondary fine waterflooding, optimize well patterns and well spacing, and identify the potential distribution of remaining oil. Physical experiments of CO_2 gravity-stabilized flooding were subsequently performed to reveal the effects of CO_2 on fluid properties and the mechanisms of enhanced oil recovery. Based on these findings, an adaptability evaluation index system was established, considering reservoir properties, reservoir conditions, fluid properties, and development parameters. The index weights were determined using the Analytic Hierarchy Process (AHP). The proposed method was applied to a typical block in the Changqing W tight oil reservoir in China. The comprehensive adaptability evaluation coefficient was calculated, and the target reservoir was classified as suitable for CO_2 gravity-stabilized flooding. Numerical reservoir simulations were further conducted to optimize key development parameters. The results suggest that the optimal conversion timing to CO_2 gravity-stabilized flooding is when the water cut reaches 60%–70%. Continuous CO_2 injection is recommended, with an injection rate of approximately 20000 m^3/d and an injection-production ratio of approximately 1.5. Based on the evaluation results, a development plan was formulated and implemented in the field. After three years of application, cumulative incremental oil production reached 642000 tons, and the staged recovery factor increased by 11 percentage points, validating the applicability and engineering significance of the proposed method. This

study establishes a scientific and systematic adaptability evaluation approach that can provide technical support and theoretical guidance for enhancing oil recovery and improving the efficiency of tight oil reservoir development.

Keywords: Tight Oil Reservoirs, Waterflooding, CO_2 Gravity-Stabilized Flooding, Secondary-Tertiary Combined Flooding, Fuzzy Comprehensive Evaluation

Introduction

Tight oil reservoirs have emerged as a key domain for the future development of global hydrocarbon resources, with technically recoverable reserves estimated at approximately 2.8 to 6.3 billion tons. These resources are primarily distributed in North America, South America, the Middle East, and several large basins in China (Liu, 2023; Luo et al., 2022a). In particular, China possesses abundant tight oil resources, with technically recoverable reserves of approximately 1.234 billion tons, mainly concentrated in the Ordos Basin, Junggar Basin, and Songliao Basin, indicating considerable development potential (Zhou et al., 2021). Tight oil reservoirs typically exhibit pronounced pore-throat heterogeneity, strong reservoir heterogeneity, and low water injectivity, which result in significant capillary forces and high flow resistance during fluid migration (Jia et al., 2024; Wang et al., 2024; Zhou et al., 2024). Conventional waterflooding in tight oil reservoirs is often constrained by high injection pressures, difficulty in establishing effective displacement fronts, and front instability, ultimately leading to low recovery efficiency (Ding et al., 2024; Pu et al., 2023).

Secondary fine waterflooding, implemented through well pattern optimization, reservoir subdivision, zonal water injection, and deep profile control using gel treatments, can partially improve reservoir utilization and enhance waterflooding performance (Bai et al., 2024; Ling et al., 2024; Wu et al., 2025). Recent studies have demonstrated that timely transition to CO_2 gravity-stabilized flooding following secondary fine waterflooding can establish a secondary-tertiary combined flooding mode. This approach effectively utilizes CO_2 to form a secondary gas cap, thereby expanding the swept volume, reducing crude oil density and viscosity, improving flow conditions, further mobilizing the remaining oil, and ultimately increasing the final recovery factor (Al-Obaidi et al., 2024; Yang et al., 2025a; Zheng et al., 2023). Field applications in medium to high-permeability reservoirs have demonstrated that fine waterflooding, enabled by optimized well pattern deployment, can be seamlessly integrated with CO_2 gravity-stabilized flooding. This combined development strategy has achieved significant improvements in water control and oil recovery, providing a mature technical basis and engineering experience for secondary-tertiary combined flooding (Al-Mudhafar and Rao, 2020; Qi et al., 2020, 2021). However, when applied to tight oil reservoirs, this approach faces considerable technical barriers due to the complexity of pore-throat structures and the poor flow connectivity of the reservoir.

Extensive research has been conducted to investigate the mechanisms and adaptability of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs. It is generally accepted that reservoir properties, reservoir conditions, fluid properties, and development parameters jointly influence displacement efficiency and production performance. In terms of reservoir properties and conditions, Ren et al. (2015) summarized typical reservoir types suitable for CO_2 gravity-stabilized flooding based on field observations and pointed out that the development potential of this method is significantly affected by geological heterogeneity and reservoir variability. Further studies focusing on pore-scale flow have revealed that poor pore-throat connectivity and pronounced heterogeneity not only limit the large-scale applicability but also have a direct impact on gas migration paths and displacement efficiency. Al-Mudhafar et al. (2018) and Luo et al. (2022b) systematically analyzed these effects through statistical analysis and reservoir simulation. Their results indicated that complex pore structures and limited connectivity exacerbate gas channeling and reduce the overall swept volume. Fluid properties are also key factors in

CO_2 flooding performance. Behnoud et al. (2023) employed pore-scale simulations to evaluate the effects of oil viscosity, interfacial tension, and fluid phase behavior on gas displacement stability and found that fluid properties play a decisive role in controlling gas flow modes. Liu et al. (2025) systematically studied the phase behavior evolution during CO_2 flooding through physical experiments and phase behavior analysis, confirming that gas expansion, light component extraction, and phase transformation are critical to improving displacement efficiency. Yang et al. (Yang et al., 2025b) further demonstrated through in-situ phase behavior monitoring and numerical simulations that phase evolution significantly affects displacement dynamics and oil recovery outcomes. In addition, development parameters such as initial reservoir pressure, recovery degree, injection timing, and injection rate also have a profound influence on the adaptability and effectiveness of CO_2 gravity-stabilized flooding. Lu et al. (Lu et al., 2023) performed sensitivity analysis and concluded that maintaining reasonable initial pressure and recovery degree is fundamental to ensuring effective displacement. Furthermore, optimizing injection timing, injection rates, and injection-production strategies can effectively delay gas breakthrough, expand the swept volume, and further enhance recovery efficiency (Al-Obaidi et al., 2022; Xiao et al., 2024; Xue et al., 2024). Overall, existing studies have preliminarily identified the key controlling factors that affect the performance of CO_2 gravity-stabilized flooding in tight oil reservoirs. These include reservoir properties, reservoir conditions, fluid properties, and development parameters, which provide a theoretical basis and parameter guidelines for adaptability evaluation of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs.

Based on the understanding of key influencing factors, further adaptability evaluation of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs can be conducted. Some researchers have systematically analyzed the adaptability of CO_2 gravity-stabilized flooding under different reservoir types, fracture structures, well patterns, and injection-production strategies through numerical reservoir simulations. These research outcomes have successfully guided field applications and achieved notable incremental oil recovery (Jadhawar and Sarma, 2012; Wo et al., 2008; Zhou et al., 2019; Zuloaga-Molero et al., 2016). However, the numerical simulation-based methods typically involve complex modeling processes and long computational cycles, making them more suitable for single-well or small-scale case predictions. These methods are often insufficient for rapid field adaptability evaluations and large-scale regional applications. In addition, some researchers have developed multi-parameter adaptability evaluation systems based on the fuzzy comprehensive evaluation method. By identifying key controlling parameters such as porosity, permeability, injection pressure, and water cut, these studies have enabled rapid and quantitative adaptability assessments of CO_2 flooding. This approach offers advantages of simplicity, high computational efficiency, and convenient field application (Chen et al., 2024; Lu et al., 2021). However, most existing statistical evaluation systems rely on single-type data sources and lack integration of dynamic and static multi-source information. Furthermore, the selection of evaluation indicators is not yet systematic, which limits the accuracy and applicability of the evaluation results. Therefore, integrating diverse dynamic and static data to establish a standardized adaptability evaluation system for secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs and developing a scientific and reliable quantitative evaluation method is considered a critical technical pathway to address the current challenges in field application.

This study proposes an adaptability evaluation method for secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs. First, based on dynamic and static monitoring data, a systematic analysis of the development performance of secondary fine waterflooding is conducted to evaluate the adaptability of well patterns, well spacing, and reservoir subdivision, and to identify the remaining oil distribution. Subsequently, physical experiments of CO_2 gravity-stabilized flooding are performed to investigate the phase behavior evolution of the CO_2–crude oil system and to elucidate the mechanisms of enhanced oil recovery. On this basis, an adaptability evaluation index system is established by comprehensively considering reservoir properties, reservoir conditions, fluid properties, and development parameters. The weight of each index is determined using the Analytic Hierarchy Process (AHP), and a fuzzy comprehensive evaluation model is applied to quantify the adaptability coefficient, thereby achieving classification of development adaptability. Based on the evaluation results, numerical reservoir simulations of CO_2 gravity-stabilized flooding are further conducted to optimize key parameters such as the conversion timing, injection mode, and injection-production rates, and to formulate field-applicable development strategies. Finally, the proposed method is applied to a typical block of the Changqing W tight oil reservoir in China to verify the accuracy and engineering feasibility of the adaptability evaluation method. The overall technical workflow is illustrated in Figure 1.

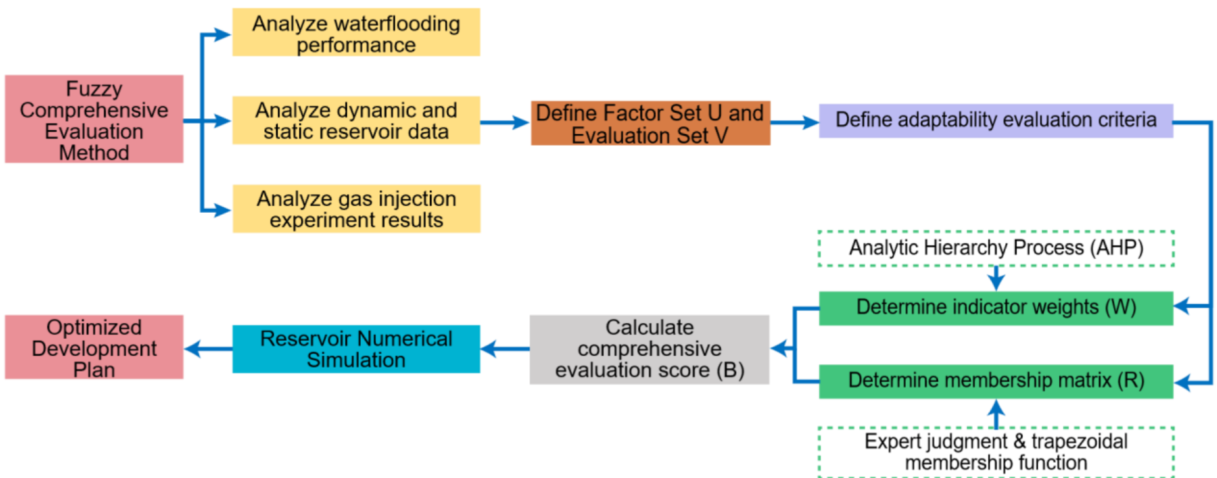

Figure 1—Workflow for adaptability evaluation and development optimization of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs.

Methodology and Evaluation Workflow

Comprehensive Evaluation of Waterflooding Performance

A scientific evaluation of waterflooding performance, identification of current development challenges, and clarification of remaining oil development potential form the foundation for the implementation of secondary-tertiary combined CO_2 gravity-stabilized flooding. In this study, the Changqing W tight oil reservoir in China was selected as the research object. Based on the water cut evolution characteristics, the waterflooding development process was divided into four stages: low water cut period (<20%), medium water cut period (20%–40%), medium-high water cut period (40%–60%), and high water cut period (>60%) (see Fig. 2).

For each stage, the production performance was analyzed by classifying oil wells according to their daily oil production and cumulative oil production. The proportion and productivity trends of different well types were statistically evaluated to reveal the dynamic characteristics and stage differences of waterflooding development. On this basis, key indicators such as water-oil ratio, recovery factor, and water cut were selected to quantitatively assess the displacement efficiency and reservoir utilization degree in the target area, and to identify injection-production imbalance zones, poorly swept zones, and remaining oil-enriched zones. Meanwhile, combined with water absorption and production profile testing data, the waterflooding patterns and water breakthrough directions of individual wells were analyzed. This analysis was used to comprehensively evaluate the rationality of well pattern deployment, the adaptability of well spacing, and the connectivity of reservoir subdivisions. Ultimately, the development challenges were thoroughly clarified, and the remaining potential distribution was systematically identified.

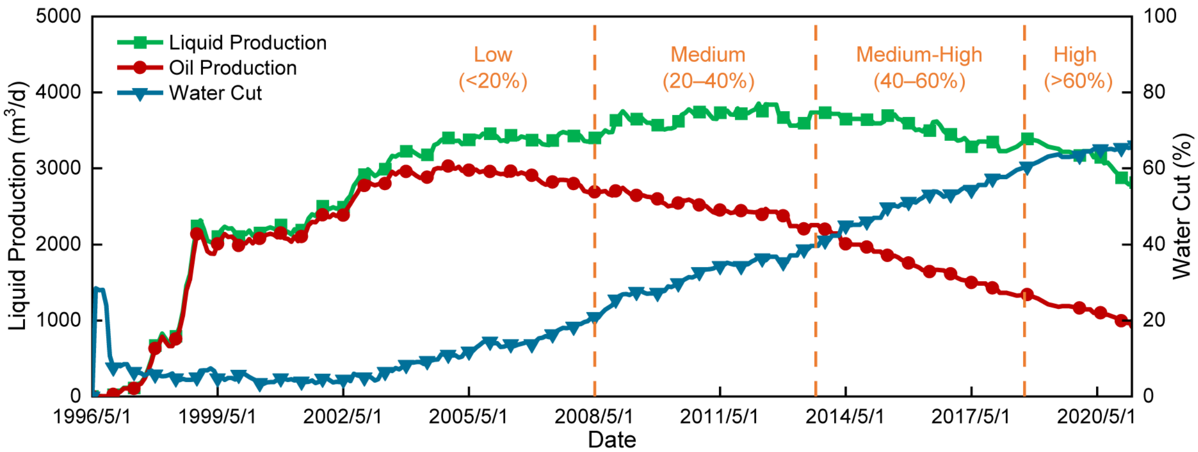

Figure 2—Classification of waterflooding development stages based on water cut evolution in the W tight oil reservoir. The waterflooding process is divided into four stages: low, medium, medium-high, and high water cut periods, representing the reservoir's production dynamics over time.

CO_2 Physical Experiments and Phase Behavior Mechanism Analysis

To analyze the phase behavior evolution and displacement mechanisms of the CO_2–crude oil system in the target reservoir and to clarify the adaptability of secondary-tertiary combined CO_2 gravity-stabilized flooding, CO_2 phase behavior and displacement experiments were conducted under reservoir conditions (55°C/17.2 MPa). The results of the expansion and differential liberation experiments (see Fig. 3) indicate that CO_2 dissolution reduces crude oil density by approximately 8.2% and viscosity by approximately 24.7%, significantly improving crude oil mobility and providing favorable conditions for gravity-stabilized flooding.

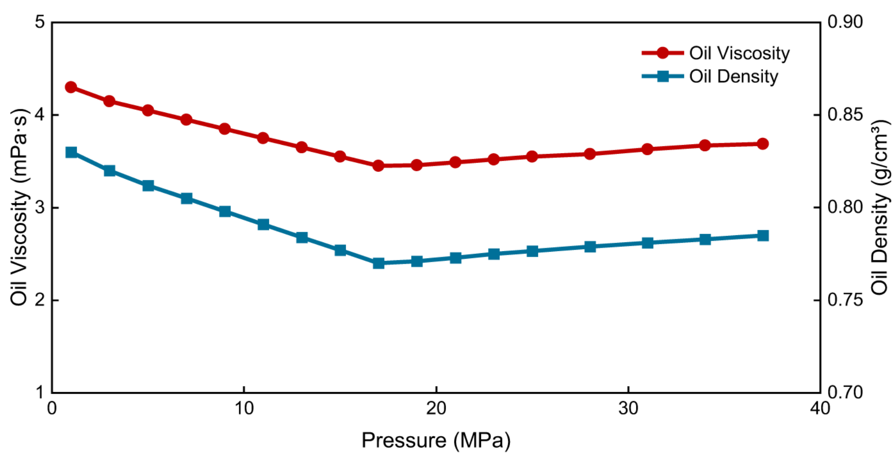

Figure 3—Variation of crude oil viscosity and density with pressure. Crude oil viscosity decreases and density significantly reduces at low pressures due to CO_2 dissolution, indicating improved crude oil mobility under increasing pressure.

Subsequently, a slim tube displacement experiment was performed to determine the minimum miscibility pressure (MMP). The results show that when the pressure increases to 16.5 MPa, the oil recovery factor rapidly rises to over 90% and then stabilizes, confirming that the MMP of the target reservoir is 16.5 MPa (see Fig. 4).

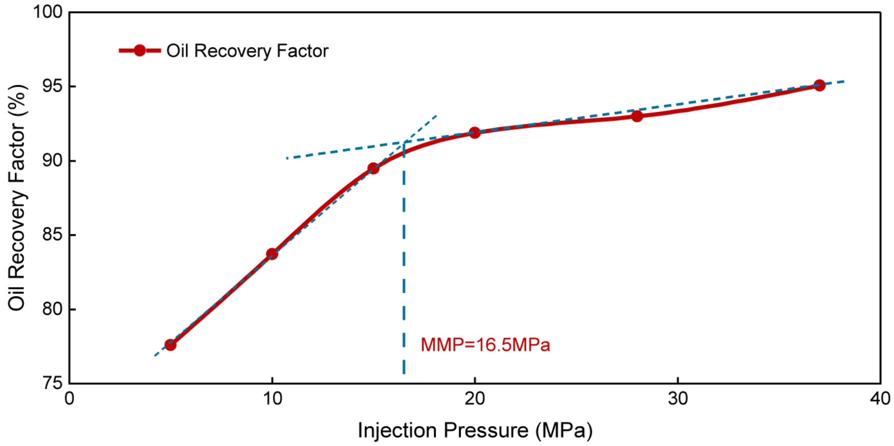

Figure 4—Determination of minimum miscibility pressure (MMP) through slim tube displacement experiment. The MMP of the target reservoir is determined to be 16.5 MPa based on the rapid increase in oil recovery factor.

Based on these findings, a physical experiment simulating the transition from waterflooding to CO_2 gravity-stabilized flooding was designed and conducted according to the similarity criterion. The displacement efficiency, oil recovery factor, and three-phase (oil–gas–water) dynamic fluid distribution characteristics at different conversion timings were compared to analyze the effect of conversion timing on water control and oil recovery enhancement. The experimental results demonstrate that reasonable control of the conversion timing helps expand the swept volume, delay the water cut rise rate, and significantly improve the ultimate oil recovery (see Fig. 5).

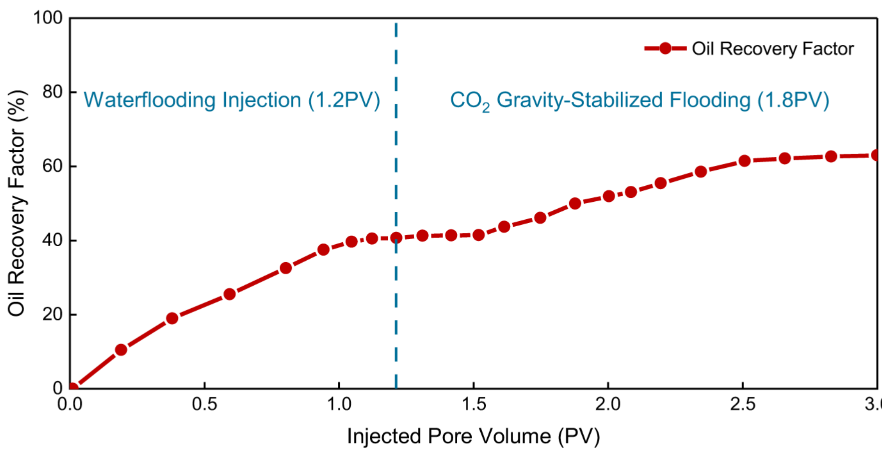

Figure 5—Oil recovery factor variation during waterflooding and subsequent CO_2 gravity-stabilized flooding. The transition point at 1.2 PV marks the shift from water injection to CO_2 injection.

Construction of the Adaptability Evaluation Index System

Based on a comprehensive understanding of waterflooding performance and CO_2–crude oil phase behavior and displacement mechanisms, multi-source data were further integrated. These data include numerical reservoir simulations, laboratory physical experiments, and field monitoring. The key factors influencing the performance of secondary-tertiary combined CO_2 gravity-stabilized flooding were systematically analyzed, and the primary controlling parameters were identified to construct the adaptability evaluation index system.

First, the sensitivity and impact of different factors on oil recovery were evaluated using visual analysis methods such as tornado charts and Pareto charts, as well as statistical methods including variance analysis and regression analysis, to identify the primary controlling factors and key sensitive parameters. Subsequently, based on experimental results from high-pressure mercury injection, core displacement, nuclear magnetic resonance (NMR), contact angle measurements, water sensitivity tests, and formation water type analyses, the roles of pore-throat structure, wettability, water sensitivity, connectivity, and fluid properties in the CO_2 displacement process were thoroughly investigated to support the construction of the evaluation index system. To ensure engineering applicability and economic feasibility, the CO_2 source-sink matching degree was incorporated into the adaptability evaluation framework.

On this basis, an evaluation index system consisting of primary indicators—namely, reservoir properties, reservoir conditions, fluid properties, and development parameters—was established based on the physical attributes and engineering features of the influencing factors. Reservoir properties primarily include parameters such as porosity, permeability, and wettability, which focus on describing the storage capacity of the rock matrix, fluid migration capability, and the stability of the displacement front. Reservoir conditions cover factors such as reservoir depth, reservoir pressure, and fracture development degree, which systematically reflect the remaining reservoir energy and interlayer connectivity. Fluid properties mainly involve the underground crude oil viscosity, density, and formation water salinity, which are used to evaluate the impact of fluid properties on development performance under reservoir conditions. Development parameters include well spacing, current recovery degree, and CO_2 source-sink matching degree, which comprehensively reflect the foundation of the well pattern, the scale of remaining oil potential, and the resource guarantee capacity for subsequent CO_2 flooding (source-sink matching). All of these parameters jointly constitute the factor set for the adaptability evaluation of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs.

Based on the identified factor set, further analyses were conducted using numerical simulations, experimental tests, and field practice to determine the reasonable value ranges of each indicator under different geological and development conditions, as well as their response characteristics to CO_2 gravity-stabilized flooding performance. According to the controlling mechanisms and the contribution of each indicator, the value ranges were classified into five evaluation grades: Excellent, Good, Fair, Poor, and Unfavorable. Ultimately, the complete adaptability evaluation index system for secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs was established, as summarized in Table 1.

Table 1—Adaptability Evaluation Index System for Secondary-Tertiary Combined CO_2 Gravity-Stabilized Flooding in Tight Oil Reservoirs.

No.	Primary Index	Secondary Index	Unit	Excellent	Good	Moderate	Poor	Very Poor
1	Reservoir Properties	Porosity	%	15–20	20–25	25–30	30–35	>35
					12–15	9–12	5–9	<5
2		Permeability	10^{-3} μm^2	<10	10–50	50–200	200–500	>500
3		Wettability	—	Oil-wet	Weakly oil-wet	Neutral	Weakly water-wet	Water-wet
4		Sub-layer thickness	m	0–8	8–16	16–24	24–32	>32
5		Pore-throat radius	μm	>0.8	0.5–0.8	0.3–0.5	0.1–0.3	<0.1
6		Water sensitivity	—	Strong	Relatively strong	Moderate	Relatively weak	Weak
7		Heterogeneity (Var. Coeff.)	—	<0.5	0.5–0.6	0.6–0.7	0.7–0.8	>0.8
8	Reservoir Conditions	Reservoir depth	m	>1500	1200–1500	1000–1200	800–1000	<800
9		Reservoir temperature	°C	>60	55–60	50–55	45–50	<45

No.	Primary Index	Secondary Index	Unit	Excellent	Good	Moderate	Poor	Very Poor
10		Reservoir pressure	MPa	>25	20–25	15–20	10–15	<10
11		Rhythm	—	Positive	Uniform	Composite	Reverse	
12		Formation dip angle	°	>20	15–20	10–15	5–10	<5
13		Fracture development degree	—	Absent	Slight	Moderate	Large	Extensive
14		Connectivity	—	Good	Relatively good	Average	Relatively poor	Poor
15	Fluid Properties	In-situ oil density	g/cm^3	<0.78	0.78–0.80	0.80–0.82	0.82–0.84	>0.84
16		In-situ oil viscosity	mPa·s	<4	4–6	6–8	8–10	>10
17		Formation water salinity	10^4 mg/L	<2	2–3	3–4	4–5	>5
18		Formation water type	—	$NaHCO_3$	$CaCl_2$	$MgCl_2$	Na_2SO_4	NaCl
19	Development Parameters	Well spacing	m	250–300	300–350 200–250	350–400 150–200	400–450 100–150	>450 <100
20		Pre-gas injection recovery factor	%	<10	10–20	20–30	30–40	>40
21		Pre-gas injection oil saturation	%	>50	40–50	30–40	20–30	<20
22		Pre-gas injection water cut	%	<85	85–90	90–93	93–95	>95
23		CO_2 source–sink matching	—	High	Relatively high	Average	Relatively low	Low

Weight Calculation

To quantify the relative importance of each evaluation indicator in the performance of secondary-tertiary combined CO_2 gravity-stabilized flooding, the Analytic Hierarchy Process (AHP) was employed for weight calculation. Based on the aforementioned research results, pairwise comparisons were conducted among indicators at the same hierarchical level within the established adaptability evaluation index system. Judgment matrices were constructed to systematically assess the influence degree of each parameter on development performance. After consistency verification ($CR < 0.1$), the local weights of each hierarchical indicator were determined (see Table 2). On this basis, the comprehensive weights of the secondary indicators were calculated by multiplying the weights of the primary indicators by the weights of their corresponding secondary indicators. This calculation ultimately established the complete weight system for subsequent adaptability evaluation.

Table 2—Weight calculation results using the Analytic Hierarchy Process (AHP).

Primary Indicator	Primary Weight	Secondary Indicator	Secondary Weight	Combined Weight
Reservoir Properties	0.33	Sub-layer thickness	0.1332	0.0440
		Permeability	0.3002	0.0991
		Porosity	0.3002	0.0991
		Heterogeneity	0.1088	0.0359
		Wettability	0.0506	0.0167
		Pore-throat radius	0.0681	0.0225
		Water sensitivity	0.0389	0.0128
Reservoir Conditions	0.33	Reservoir depth	0.0613	0.0202
		Formation dip angle	0.0640	0.0211
		Reservoir temperature	0.2313	0.0763
		Reservoir pressure	0.2313	0.0763
		Fracture development degree	0.0721	0.0238
		Rhythm	0.0325	0.0107
		Connectivity	0.3075	0.1015
Fluid Properties	0.2	Formation water salinity	0.1044	0.0209
		In-situ oil density	0.4225	0.0845
		In-situ oil viscosity	0.4225	0.0845
		Formation water type	0.0506	0.0101
Development Parameters	0.14	Well spacing	0.0370	0.0052
		Pre-gas injection recovery factor	0.1420	0.0199
		Pre-gas injection water cut	0.0717	0.0100
		Pre-gas injection oil saturation	0.2791	0.0391
		CO_2 source–sink matching	0.4702	0.0658

Comprehensive Adaptability Evaluation

Based on the established evaluation index system and corresponding weights, the measured values of each indicator were classified into the predefined evaluation grade intervals according to the preset classification criteria. Subsequently, the membership function method was employed to calculate the membership degree of each evaluation grade, and a single-factor membership matrix **R** was constructed to quantify the influence degree of each indicator on the adaptability of secondary-tertiary combined CO_2 gravity-stabilized flooding:

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \tag{1}$$

where m is the number of secondary indicators, n is the number of evaluation grades, and r_{mn} represents the membership degree of the m^{th} with respect to the n^{th} evaluation grade.

Based on the comprehensive weights of the secondary indicators, the weighted average method was used to perform fuzzy operation and calculate the comprehensive membership degree vector **B** for each evaluation grade:

$$\mathbf{B} = \mathbf{W} \cdot \mathbf{R} \tag{2}$$

where $\mathbf{B} = \{b_1, b_2, \ldots, b_n\}$ represents the comprehensive membership degree at the n^{th} evaluation grade.

Finally, according to the principle of maximum membership degree, the evaluation grade with the highest membership degree in **B** was selected as the final adaptability evaluation result for the target tight oil reservoir in secondary-tertiary combined CO_2 gravity-stabilized flooding.

Development Adjustment Plan

Based on the comprehensive adaptability evaluation results, targeted development adjustment plans were further designed for tight oil reservoirs classified as "Excellent" and "Good" in the evaluation. Using the existing reservoir matrix model and integrating drilling, logging, and hydraulic fracturing data, artificial fracture modeling was performed. Fracture parameters were inversely derived by fitting the construction data to accurately characterize the spatial distribution of fractures. Subsequently, the fracture model was coupled with the matrix model using a superposition algorithm to construct a multi-scale fine geological model of the reservoir.

On this basis, the static parameters of the geological model were imported into the reservoir simulation software. Combined with relative permeability curves, production performance, and PVT data calibrated through phase behavior experiments, a reservoir numerical simulation model was established (see Fig. 6). Production history matching was conducted to ensure the reliability of the model.

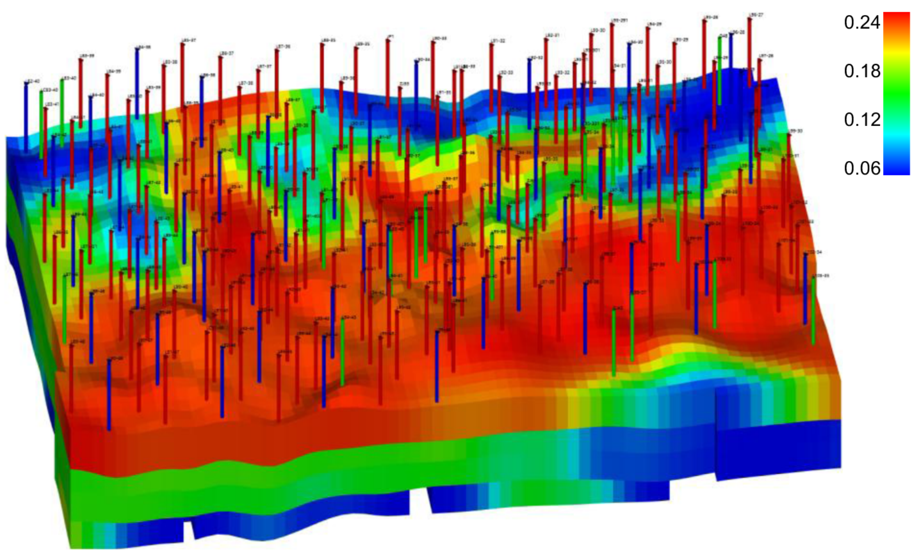

Figure 6—Numerical simulation model and porosity distribution of the CO_2 gravity-stabilized flooding reservoir.

Based on the established reservoir simulation model, the development adjustment plan for secondary-tertiary combined CO_2 gravity-stabilized flooding was further optimized. By setting different parameter combinations, the effects of key technical parameters—including well pattern and spacing, conversion timing, injection method, and injection-production rates—on production dynamics and oil recovery were systematically analyzed. The water control and oil recovery enhancement effects under different parameter combinations were evaluated by comparing the dynamic trends of daily liquid production, daily oil production, and water cut, in combination with the evolution characteristics of the saturation and pressure fields. Finally, an optimized injection-production scheme was selected to form a targeted development adjustment plan for the target block, providing technical support for subsequent field development adjustments.

Case Study

Reservoir Overview

This section provides a brief overview of the geological and development background of the W reservoir. In this study, the W tight oil reservoir in the Changqing Oilfield was selected as the research target. The established adaptability evaluation method was systematically applied to assess the feasibility of secondary-tertiary combined CO_2 gravity-stabilized flooding. The W reservoir is a typical tight oil reservoir characterized by strong heterogeneity, complex pore-throat structures, and pronounced water sensitivity.

Development trials for the reservoir commenced in 1996. In 1997, a reverse nine-spot well pattern was implemented for waterflooding development. Full-scale development began in 1998, and by 2002, the reservoir had been developed into an integrated oilfield with an annual oil production exceeding 600000 tons. In 2004, the reservoir as a whole entered the production decline stage, and by 2014, the water cut exceeded 60%, marking the transition into the high water cut development stage. During the long-term waterflooding process, localized high-pressure zones persisted, the water cut increased rapidly, and overall development performance gradually declined. Based on the waterflooding performance evaluation and fluid dynamic evolution from reservoir numerical simulation, it was observed that the remaining oil is locally enriched in the plane and predominantly distributed in the upper sections of the reservoir in the vertical profile. Currently, the reservoir faces increasingly severe issues of insufficient areal sweep and difficulties in vertical adjustment, necessitating the exploration of effective enhanced oil recovery technologies.

Preliminary phase behavior analysis and physical property experiments indicated that under the current reservoir pressure-temperature conditions, CO_2 and crude oil can achieve miscibility, which can effectively enhance fluid mobility and expand the swept volume. Meanwhile, due to the relatively high density of CO_2, gravity segregation tends to occur during the displacement process. The synergistic effect of miscibility and gravity segregation can effectively mitigate the current development challenges of insufficient areal sweep and poor vertical displacement efficiency, providing a solid physical basis for the implementation of secondary-tertiary combined CO_2 gravity-stabilized flooding. The comprehensive analysis confirms that the W reservoir possesses the technical feasibility and application potential for secondary-tertiary combined CO_2 gravity-stabilized flooding.

Adaptability Evaluation Results

Based on the actual dynamic and static parameters of the W reservoir, the adaptability evaluation method for secondary-tertiary combined CO_2 gravity-stabilized flooding established in this study was applied. According to the predefined classification criteria, the membership degrees of each evaluation indicator were determined (see Table 3). On this basis, combined with the comprehensive weights of each indicator, the adaptability score was calculated using the fuzzy comprehensive evaluation method.

Table 3—Basic reservoir data and membership degrees for CO_2 gravity-stabilized flooding in the Changqing W reservoir. The membership degrees were determined based on the predefined classification criteria.

Primary Indicator	Secondary Indicator	Unit	Weight	Reservoir Value	Excellent	Good	Moderate	Poor	Very Poor
Reservoir Properties	Porosity	%	0.0991	12.69	0.23	1	0.77	0	0
	Permeability	10^{-3} μm^2	0.0991	1.81	1	0.181	0	0	0
	Wettability	—	0.0167	Neutral	0	0	1	0	0
	Sub-layer thickness	m	0.044	12.2	0.475	1	0.525	0	0
	Pore-throat radius	μm	0.0225	0.186	0	0	0.43	1	0.57

Primary Indicator	Secondary Indicator	Unit	Weight	Reservoir Value	Excellent	Good	Moderate	Poor	Very Poor
	Water sensitivity	—	0.0128	Relatively strong	0	1	0	0	0
	Heterogeneity coefficient	—	0.0359	0.25	1	0.5	0	0	0
Fluid Properties	In-situ oil density	g/cm^3	0.0843	0.794	0.3	1	0.7	0	0
	In-situ oil viscosity	mPa's	0.0843	1.95	1	0	0	0	0
	Formation water salinity	10^4 mg/L	0.0208	8.22	0	0	0	0	1
	Formation water type	—	0.0101	$CaCl_2$-type	1	0	0	0	0
Reservoir Conditions	Reservoir depth	m	0.0202	1850	1	0	0	0	0
	Reservoir temperature	°C	0.0763	54.73	0	0.946	1	0.054	0
	Reservoir pressure	MPa	0.0763	17.2	0	0.44	1	0.56	0
	Rhythm	—	0.0107	Positive	1	0	0	0	0
	Formation dip angle	°	0.0211	1	0	0	0	0.2	1
	Fracture development	—	0.0238	Moderate	0	0	1	0	0
	Connectivity	—	0.1015	Relatively good	0	1	0	0	0
Development Parameters	Well spacing	m	0.0052	330	0.4	1	0.6	0	0
	Pre-gas injection recovery	%	0.0199	22.6	0	0.74	1	0.26	0
	Pre-gas injection oil saturation	%	0.0392	47	0.7	1	0.3	0	0
	Pre-gas injection water cut	%	0.0101	67.2	1	0	0	0	0
	CO_2 source–sink matching	—	0.066	Relatively high	0	1	0	0	0

Based on the adaptability evaluation results presented in Table 4, and following the principle of maximum membership degree, the adaptability evaluation grade of the W reservoir was determined to be "Excellent." This indicates that the reservoir demonstrates favorable adaptability for secondary-tertiary combined CO_2 gravity-stabilized flooding. It is recommended to implement CO_2 gravity-stabilized flooding development in this reservoir.

Table 4—Comprehensive adaptability evaluation results of secondary-tertiary combined CO_2 gravity-stabilized flooding in the Changqing W reservoir.

Evaluation Level	Excellent	Good	Moderate	Poor	Very Poor
Membership Value	0.2918	0.244	0.2246	0.0719	0.1677

Development Plan Determination

Based on the adaptability evaluation results, reservoir numerical simulation of secondary-tertiary combined CO_2 gravity-stabilized flooding was conducted for the W reservoir to determine the development adjustment plan. The optimization results of key parameters—including conversion timing, injection method, injection rate, and injection-production ratio—are shown in Fig. 7. It can be observed that these key parameters have a significant impact on the effectiveness of CO_2 gravity-stabilized flooding. Specifically, converting to CO_2 gravity-stabilized flooding when the water cut reaches 60%–70% during waterflooding development can effectively delay the water cut rise, expand the swept volume, and enhance oil recovery. Continuous CO_2 injection is superior to cyclic injection and water-alternating-gas (WAG) injection. Increasing the gas injection rate can rapidly compensate for reservoir pressure depletion and improve development performance; however, when the injection rate exceeds 20000 m^3/d, gas channeling intensifies, resulting in diminished development performance. The injection-production ratio is recommended to be maintained at approximately 1.5 to balance displacement efficiency and the stability of the CO_2 flooding process.

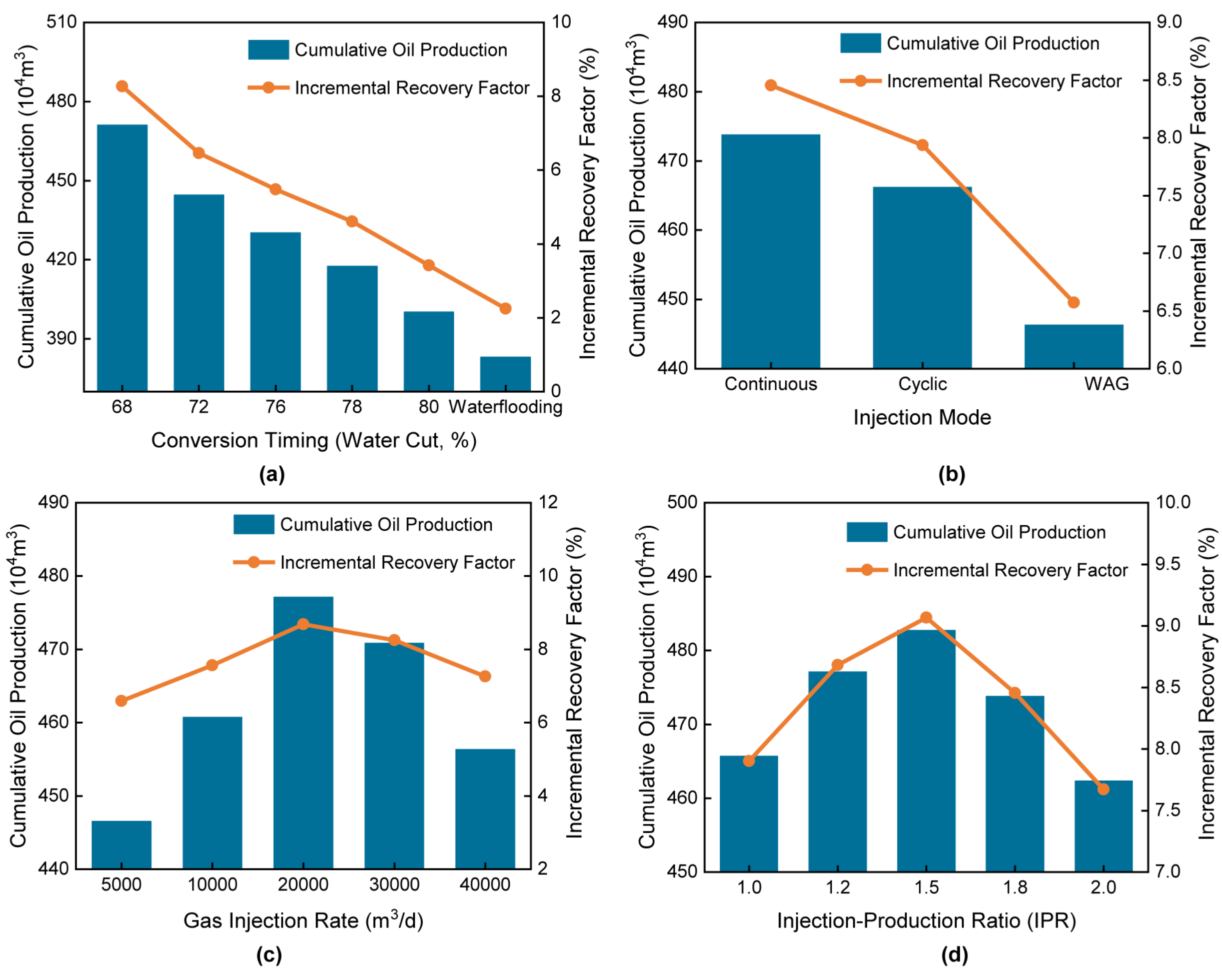

Figure 7—Optimization results of the development plan for secondary-tertiary combined CO_2 gravity-stabilized flooding. (a) CO_2 conversion timing; (b) Injection method; (c) Gas injection rate; (d) Injection-production ratio optimization.

Based on the numerical simulation optimization results, a secondary-tertiary combined CO_2 gravity-stabilized flooding development plan suitable for the W reservoir was ultimately formulated. A field pilot test of secondary-tertiary combined CO_2 gravity-stabilized flooding was conducted in a typical block of the W reservoir. Field dynamic monitoring results indicated that, after the implementation of secondary-tertiary combined CO_2 gravity-stabilized flooding, the sweep efficiency of the injection profile was significantly improved, the water cut rise rate was notably reduced, and the utilization of remaining oil was effectively enhanced. Compared with conventional waterflooding development, the cumulative incremental oil production reached 642000 tons over three years, and the staged recovery factor increased by 11 percentage points. The field application validated the adaptability and effectiveness of the secondary-tertiary combined CO_2 gravity-stabilized flooding technology in tight oil reservoirs, providing technical support for future large-scale field applications.

Conclusion

To address the limitations of existing adaptability evaluation methods for tight oil reservoirs, such as single evaluation approaches and large evaluation errors, this study systematically developed and applied an adaptability evaluation method for secondary-tertiary combined CO_2 gravity-stabilized flooding. The main conclusions are as follows:

1. Taking a typical block of the W tight oil reservoir in the Changqing Oilfield as the research target, the waterflooding development process was divided and the development performance was evaluated to clarify the current development challenges and remaining oil distribution characteristics. The results show that, in the late stage of waterflooding, the remaining oil is primarily enriched in poorly swept zones between wells in the plane and mainly retained in the upper sections of the reservoir in the vertical profile. The reservoir faces problems such as insufficient areal sweep, poor vertical displacement efficiency, and injection-production imbalance, which urgently require development mode optimization to further improve oil recovery.
2. Through CO_2 phase behavior analysis and physical simulation experiments of gravity-stabilized flooding, the phase evolution characteristics of the CO_2–crude oil system and the oil displacement mechanisms were revealed. The results indicate that miscibility between CO_2 and crude oil can be achieved under reservoir conditions in the target reservoir. CO_2 injection can significantly reduce crude oil viscosity and density, improving fluid mobility and providing favorable conditions for gravity-stabilized flooding. The comprehensive evaluation demonstrates that the W tight oil reservoir possesses the physical basis and application potential for transitioning from waterflooding to CO_2 gravity-stabilized flooding.
3. Based on dynamic analysis, experimental studies, reservoir numerical simulation, and physical property testing, key controlling indicators were identified by comprehensively integrating reservoir properties, reservoir conditions, fluid properties, and development parameters. An adaptability evaluation index system for secondary-tertiary combined CO_2 gravity-stabilized flooding was established. The indicator weights were determined using the Analytic Hierarchy Process (AHP), and the membership degrees of individual indicators were calculated using membership functions. A fuzzy comprehensive evaluation model was constructed to enable rapid, quantitative, and accurate adaptability evaluation of secondary-tertiary combined CO_2 gravity-stabilized flooding in tight oil reservoirs.
4. The proposed evaluation method was applied to a typical block of the W tight oil reservoir to conduct adaptability evaluation and formulate a development adjustment plan. The results indicate that the target reservoir is suitable for secondary-tertiary combined CO_2 gravity-stabilized flooding. It is recommended to convert to CO_2 gravity-stabilized flooding when the water cut reaches 60%–70%,

adopt continuous CO_2 injection, maintain a gas injection rate of approximately 20000 m^3/d, and keep the injection-production ratio at around 1.5.

5. A field pilot test was conducted based on the optimized development plan. The results show that, after converting from waterflooding to CO_2 gravity-stabilized flooding, the cumulative incremental oil production reached 642000 tons over three years, and the staged recovery factor increased by 11 percentage points. The field application validated the adaptability and effectiveness of the secondary-tertiary combined CO_2 gravity-stabilized flooding method in tight oil reservoirs and further demonstrated the practicality and applicability of the proposed evaluation method, providing theoretical support and technical guidance for the efficient development of tight oil reservoirs.

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References

Al-Mudhafar, W.J., Rao, D.N., and Srinivasan, S.2018. Reservoir sensitivity analysis for heterogeneity and anisotropy effects quantification through the cyclic CO_2-assisted gravity drainage EOR process – a case study from South Rumaila oil field. *Fuel* **221**: 455–468. https://doi.org/10.1016/j.fuel.2018.02.121

Al-Mudhafar, W.J., and Rao, D.N.2020. Potential application of the CO_2-assisted gravity drainage process in a mature oil field: insights from reservoir-scale EOR evaluation. *International Journal of Oil, Gas and Coal Technology* **25** (1): 19–40. https://doi.org/10.1504/IJOGCT.2020.108852

Al-Obaidi, D.A., Al-Mudhafar, W.J., and Al-Jawad, M.S.2022. Experimental evaluation of carbon dioxide-assisted gravity drainage process (CO_2-AGD) to improve oil recovery in reservoirs with strong water drive. *Fuel* **324**: 124409. https://doi.org/10.1016/j.fuel.2022.124409

Al-Obaidi, D.A., Al-Mudhafar, W.J., Hussein, H.H., and Rao, D.N.2024. Experimental influence assessments of water drive and gas breakthrough through the CO_2-assisted gravity drainage process in reservoirs with strong aquifers. *Fuel* **370**: 131873. https://doi.org/10.1016/j.fuel.2024.131873

Bai, Y.Y., Pu, W.F., Jin, X., Shen, C., and Ren, H.L.2024. Review of the micro and macro mechanisms of gel-based plugging agents for enhancing oil recovery of unconventional water flooding oil reservoirs. *Journal of Molecular Liquids* **399**: 124318. https://doi.org/10.1016/j.molliq.2024.124318

Behnoud, P., Khorsand Movaghar, M.R., and Sabooniha, E.2023. Numerical analysis of pore-scale CO_2-EOR at near-miscible flow condition to perceive the displacement mechanism. *Scientific Reports* **13** (1): 1–16. https://doi.org/10.1038/s41598-023-39706-1

Chen, B., Li, H., Zhu, Q., Zhai, D., Xiao, M., Chen, W., Wang, W., and Dong, L.2024. Study on fuzzy comprehensive evaluation of influencing factors during CO_2 flooding. Presented at the 11th International Conference on Energy Engineering and Environmental Engineering, Xiamen, China, 17-18 August. https://doi.org/10.1007/978-3-031-76208-6_13

Ding, C., Chen, J., Yang, G., Bao, R.F., Dou, Y., and Song, K.P.2024. Novel method for the rapid evaluation of pressure depletion in tight oil reservoirs. *Advances in Geo-energy Research* **11** (1): 74–80. https://doi.org/10.46690/ager.2024.01.07

Jadhawar, P.S., and Sarma, H.K.2012. Effect of well pattern and injection well type on the CO_2-assisted gravity drainage enhanced oil recovery. *Journal of Petroleum Science & Engineering* **98**: 83–94. https://doi.org/10.1016/j.petrol.2012.09.004

Jia, N.H., Lv, W., Liu, Q.J., Wang, D.G., Liu, F.Z., and Hu, Z.2024. Pore-scale modeling of pressure-driven flow and spontaneous imbibition in fracturing-shut-In-flowback process of tight oil reservoirs. *International Journal of Energy Research* **2024**: 3505763. https://doi.org/10.1155/2024/3505763

Ling, Y.G., Wang, X.Q., Gao, X., Zhu, Y.L., Wang, R.H., Wei, T., and Yin, S.2024. Injection production adjustment technology for ultra-low permeability reservoirs in the ordos basin. *Science Technology and Engineering* **24** (26): 11102–11110 (in Chinese).

Liu, S.Y.2023. Review of the development status and technology of tight oil: advances and outlook. *Energy & Fuels* **37** (19): 14645–14665. https://doi.org/10.1021/acs.energyfuels.3c02726

Liu, X.L., Chen, H., Li, Y., Zhu, Y.W., Liao, H.Y., Zhao, Q.M., Zhou, X.M., and Zeng, H.B.2025. Oil production characteristics and CO_2 storage mechanisms of CO_2 flooding in ultra-low permeability sandstone oil reservoirs. *Petroleum Exploration and Development* **52** (1): 196–207. https://doi.org/10.1016/S1876-3804(25)60014-0

Lu, Y., Liu, R., Wang, K., Tang, Y., and Cao, Y.2021. A study on the fuzzy evaluation system of carbon dioxide flooding technology. *Energy Science & Engineering* **9** (2): 239–255. https://doi.org/10.1002/ese3.844

Lu, H., Xin, X., Ye, J., and Yu, G.2023. Analysis of factors impacting CO_2 assisted gravity drainage in oil reservoirs with bottom water. *Processes* **11** (12): 3290. https://doi.org/10.3390/pr11123290

Luo, Q., Gao, Y., Zhang, Z.Y., Wang, S.C., Hong, L., Ma, W., and Xu, Q.2022a. A comparative study of geological conditions of tight oils in China and USA. *Petroleum Geology & Experiment* **44** (2): 199–209 (in Chinese). https://doi.org/10.11781/sysydz202202199

Luo, J.S., Hou, Z.M., Feng, G.Q., Liao, J.X., Haris, M., and Xiong, Y.2022b. Effect of reservoir heterogeneity on CO_2 flooding in tight oil reservoirs. *Energies* **15** (9): 3015. https://doi.org/10.3390/en15093015

Pu, C.S., Kang, S.F., Pu, J.Y., Gu, X.Y., Gao, Z.D., Wang, Y.D., and Wang, K.2023. Progress and development trend of water huff-n-puff technology for horizontal wells in tight oil reservoirs in China. *Acta Petrolei Sinica* **44** (1): 188–206 (in Chinese). https://doi.org/10.7623/syxb202301012

Qi, Z.Y., Liu, T., Xi, C.F., Zhang, Y.J., Shen, D.H., Mu, H., Dong, H., Zheng, A.P., Yu, K.Q., Li, X.L., Jiang, Y.W., Wang, H.Z., Li, H.Z., and Babadagli, T.2020. A field pilot test on CO_2 assisted steam-flooding in a steam-flooded heavy oil reservoir in China. Presented at the SPE Russian Petroleum Technology Conference, Moscow, Russia, 12-14 October. SPE-201832-MS. https://doi.org/10.2118/201832-MS

Qi, Z.Y., Liu, T., Xi, C.F., Zhang, Y.J., Shen, D.H., Mu, H., Dong, H., Zheng, A.P., Yu, K.Q., Li, X.L., Jiang, Y.W., Wang, H.Z., Babadagli, T., and Li, H.Z.2021. Status quo of a CO_2-assisted steam-flooding pilot test in China. *Geofluids* **2021**: 9968497. https://doi.org/10.1155/2021/9968497

Ren, B., Zhang, L., Huang, H., Ren, S., Chen, G., and Zhang, H.2015. Performance evaluation and mechanisms study of near-miscible CO_2 flooding in a tight oil reservoir of jilin oilfield china. *Journal of Natural Gas Science and Engineering* **27**: 1796–1805. https://doi.org/10.1016/j.jngse.2015.11.005

Wang, Z.Z., Zhang, Z., Liu, Y.J., Wei, B., and Lin, X.H.2024. Experimental study on porosity & permeability characteristics of typical tight oil reservoirs. *Chemistry and Technology of Fuels and Oils* **60** (1): 80–87. https://doi.org/10.1007/s10553-024-01659-0

Wo, S., Yin, P., Blakeney-DeJarnett, B., and Mullen, C.E.2008. Simulation evaluation of gravity stable CO_2 flooding in the muddy reservoir at Grieve Field, Wyoming. Presented at the SPE Symposium on Improved Oil Recovery, Tulsa, Oklahoma, USA, 19-23 April. SPE-113482-MS. https://doi.org/10.2118/113482-MS

Wu, W.S., Song, G.X., Zhang, H., Wang, X.K., and Song, Z.J.2025. Optimization of well spacing with an integrated workflow: a case study of the Fuyu tight oil reservoir in the Daqing oil field, China. *Processes* **13** (4): 1008. https://doi.org/10.3390/pr13041008

Xiao, P.F., Cui, M.L., Zhang, W., Hu, Q.C., Zhao, S.X., Wang, R., and Tang, Y.Q.2024. Experimental and dimensional analysis of CO_2-assisted gravity drainage in low permeability dip reservoirs of east China. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects* **46** (1): 7972–7982. https://doi.org/10.1080/15567036.2020.1763519

Xue, J.J., Gao, H., Ma, Z.G., Shi, H.Q., Li, X.L., Li, T., Cheng, Z.L., Wang, C., Li, P., and Zhang, N.2024. Analyzing the microscopic production characteristics of CO_2 flooding after water flooding in tight oil sandstone reservoirs utilizing NMR and microscopic visualization apparatus. *Atmosphere* **15** (4): 487. https://doi.org/10.3390/atmos15040487

Yang, L., Liu, Y., Wang, W.D., Li, M.J., Wang, S.L., Xu, B.C., Shi, Y.M., and Chen, H.2025a. Pore-scale flow mechanism of immiscible gas-assisted gravity drainage in strongly heterogeneous glutenite reservoirs. *International Journal of Coal Science & Technology* **12** (1): 12. https://doi.org/10.1007/s40789-025-00758-5

Yang, H., Wang, H., Wang, X.Z., Xin, Y.C., He, L.J., Jiao, Z.S., and Zou, J.2025b. Experimental and modeling assessment of CO_2 EOR and storage performances in tight oil reservoir, yanchang oilfield, china. *Journal of CO2 Utilization* **97**: 103125. https://doi.org/10.1016/j.jcou.2025.103125

Zheng, W.K., Li, Z.Y., Huang, H., Cao, X.P., Wu, G.C., Wang, J., and Kang, H.2023. Evaluation and improvement of CO_2 assisted gravity drainage for enhanced oil production in China. *Energy Reports* **10**: 4465–4473. https://doi.org/10.1016/j.egyr.2023.10.090

Zhou, X., Yuan, Q., Zhang, Y., Wang, H., Zeng, F., and Zhang, L.2019. Performance evaluation of CO_2 flooding process in tight oil reservoir via experimental and numerical simulation studies. *Fuel* **236**: 730–746. https://doi.org/10.1016/j.fuel.2018.09.035

Zhou, N.W., Lu, S.F., Wang, M., Huang, W.B., Xiao, D.S., Jiao, C.X., Wang, J.M., Tian, W.C., Zhou, L., Chen, F.W., Liu, W., and Wang, Z.X.2021. Limits and grading evaluation criteria of tight oil reservoirs in typical continental basins of China. *Petroleum Exploration and Development* **48** (5): 1089–1100. https://doi.org/10.1016/S1876-3804(21)60093-9

Zhou, X.F., Wei, J.G., Zhao, J.F., Zhang, X.Y., Fu, X.F., Shamil, S., Abdumalik, G., Chen, Y.H., and Wang, J.2024. Study on pore structure and permeability sensitivity of tight oil reservoirs. *Energy* **288**: 129632. https://doi.org/10.1016/j.energy.2023.129632

Zuloaga-Molero, P., Yu, W., Xu, Y., Sepehrnoori, K., and Li, B.2016. Simulation study of CO_2-EOR in tight oil reservoirs with complex fracture geometries. *Scientific Reports* **6** (1): 33445. https://doi.org/10.1038/srep33445